Bogleheads

Dollar-cost averaging

(Redirected from Dollar cost averaging)

Dollar-cost averaging (DCA) means dividing an available investment lump sum into equal parts, and then periodically investing each part.[note 1] DCA is an alternative to lump sum investing, which is to make the entire investment immediately. It is a technique to overcome fear in investing by mitigating the risk of loss over the short term.

For any sequence of constant-dollar purchases[note 2] of an asset whose price fluctuates, you would purchase more shares when prices are lower than when prices are high.[1] As a result, the average cost per share will be lower than the average price per share over the same period.

This consequence of DCA provides an argument to persuade investors to make systematic, periodic fixed dollar amount investments regardless of market prices.

The term DCA is also encompasses similar investment concepts such as periodic automatic investment (used almost universally by individual investors to fund retirement accounts out of earned income).

A variant on the idea of DCA is "reverse dollar-cost averaging", which relates to periodic automatic withdrawals made during retirement.[2] Also, value averaging closely resembles dollar-cost averaging but differs in that you select a target growth rate or amount on their asset base or portfolio each period, and then adjusts the next period's contribution according to the relative gain or shortfall made on the original asset base.[3]

Dollar-cost averaging versus lump sum

When you are ready to invest money, a common question is whether you should invest it as a **lump sum** or **dollar-cost average (DCA)** by splitting your investment across several payments. The answer depends on the degree to which you are willing to accept lower expected returns in exchange for lower potential losses (aversion to possible loss).

In most cases, you are moving your money from cash (or the equivalent, a low-yielding money market) to some mix of stocks and bonds. The expected return of both stocks and bonds is higher than cash. However, their volatility is higher as well. The risk is that just after making your investment, the market could crash, causing your investment to quickly lose value due to the unfortunate timing of your purchase.

Of course, according to Bogleheads investment philosophy, you should only be investing in the first place into a diversified asset allocation. Also, you should only be holding volatile funds like stocks and bonds if your investing horizon is long enough to ride out their volatility.

Lump sum investing will always carry a higher expected return, because it immediately moves your funds from asset classes with lower expected returns to ones with higher expected returns. Note that higher *expected* returns do not guarantee that your *actual* returns will be higher. A Vanguard study[4]

concluded that lump sum investing has historically produced higher returns two thirds of the time when the DCA plan was implemented over 12 months. Lump sum was also more favorable when the analysis used risk-adjusted returns.[4] DCA plans with longer timelines have worse results. A related Vanguard publication notes that if a DCA plan was implemented over 36 months, lump sum investing produced higher returns approximately 92% of the time. In addition, over a 12-month period, the maximum downside potential of DCA was higher than the maximum upside potential.[5]

Some investors are willing to sacrifice some expected return to reduce their potential loss, knowing that higher expected return comes with higher potential loss. For those investors, dollar-cost averaging is superior because it reduces the chances of investing all the money just before a market drop. Instead, as the price fluctuates each month, you will buy more shares when the price is low and less when it is high. In such volatile markets your actual return can be higher than the return of the lumpsum which will match the return of the assets.

Many new investors are more interested in minimizing their potential loss, and it is important that an ill-timed market drop does not scare them off from investing in the future. Many experienced investors are more interested in maximizing their expected returns. You can also decide to split the difference, where you invest half immediately and the other half over 6 or so months.

Costs

Most no-load mutual funds allow small, regular investments with no fees. However, most investors pay a commission to purchase a load-fund or an Exchange-traded fund (ETF). Depending on the size of the commission and the investment, dollar-cost averaging into a load-fund or an ETF could be far more expensive than purchasing as a lump sum. For ETFs, the bid/ask spread is the same between DCA and lump sum.

Automatic investment

Most investors make regular contributions through their 401(k) plans or by having a set amount auto-deducted from their bank account into an IRA or taxable account. When this money is automatically invested, it has the same benefit of dollar-cost averaging that you buy more shares when prices are low and less when they are high. However, this form of investing is not dollar-cost averaging. It is called **periodic investing**. The difference is that periodic investing is maximizing expected return, because you are investing the money as soon as you have it. DCA applies when you have the money to invest, but delay doing so.

Switching to a more conservative portfolio

Some investors who are new to a Bogleheads investment philosophy may find their portfolios contain too much equity. Even worse, some investors have a large portion of their net worth held in a single stock, such as through an employee purchase plan. They generally want to sell some of their equities and buy bonds. If you want to move from high risk and volatility assets to lower risk and volatility

assets, a lump sum transaction makes more sense than DCA. This is because lump sum both immediately moves to the expected return and volatility you want, while also minimizing potential regret.

Reverse dollar-cost averaging

Investors who regularly add to their investment portfolios with **periodic investing** purchase more shares when prices are low and less shares when prices are high. Author Henry K. Hebeler[6] points out that investors **periodically distributing** retirement income from their portfolios experience the reverse action, redeeming more shares when prices are low and less shares when prices are high. Hebeler terms this process, "reverse dollar-cost averaging" with results that should, on average, reduce expected returns.

Examining rolling 20 year investment periods from 1927 to 1995 for a portfolio consisting of 50% large cap stocks (S&P 500); 40% long term corporate bonds; and 10% treasury bills, with expense loadings of 1.50% for stocks; 0.50% for bonds; and 0.30% for cash), Hebeler provides average real returns for four scenarios: the portfolio; the accumulating investor (saving for retirement); the distributing investor (withdrawal during retirement); as well as the results of 80% percentile returns (reflecting desires for portfolio survivability for retirement distributions). Indexing the portfolios would add 0.70% to each portfolio's returns simply due to reduced costs.[2]

Average real returns (rolling 20 years) 1927-1995

	Portfolio return 50% stock/40% bond/10% cash	**Accumulating investor**	**Distributing investor**
Portfolio	2.9%	3.2%	2.6%
80 percentile	1.2%	0.8%	0.3%

Notes

1. The origin of the term **Dollar cost averaging** (DCA) is the distinction between buying a fixed number of shares every period and investing a fixed dollar amount every period. Over time the meaning has been extended.
2. Equal parts means that the amount (dollars) is constant, hence the term "constant-dollar".

See also

- Value averaging
- Lump sum vs pension

References

1. "Dollar-Cost Averaging (DCA) Explained With Examples and Considerations" (https://www.investopedia.com/terms/d/dollarcostaveraging.asp). Investopedia. April 26, 2023. Retrieved August 12, 2023.
2. Henry K. Hebeler (January 2, 2001). "Reverse Dollar Cost Averaging" (http://www.analyzenow.com/Articles/Investments/Investment%20Articles/ReverseDollarCostAvg1-1-01.pdf) (PDF). Retrieved

August 12, 2023.
3. "Value Averaging: What it Means, Examples" (https://www.investopedia.com/terms/v/value_averaging.asp). Investopedia. May 5, 2022. Retrieved August 12, 2023.
4. "Dollar-cost averaging just means taking risk later" (https://web.archive.org/web/20121107195530/https://pressroom.vanguard.com/nonindexed/7.23.2012_Dollar-cost_Averaging.pdf) (PDF). Vanguard. July 23, 2012. Archived from the original (https://pressroom.vanguard.com/nonindexed/7.23.2012_Dollar-cost_Averaging.pdf) (PDF) on November 7, 2012.
5. Daniel B. Berkowitz (2016). "Invest now or temporarily hold your cash?" (https://www.optimizedportfolio.com/wp-content/uploads/2021/03/vanguard-invest-now-or-temporarily-hold-your-cash.pdf) (PDF). Vanguard.
6. Henry K. Hebeler (April 27, 2007). *Getting Started in A Financially Secure Retirement*. John Wiley & Sons. ISBN 978-0-470-11778-1.

External links

- A behavioral framework for dollar-cost averaging: Dollar-cost averaging may not be rational behavior, but it is perfectly normal behavior (https://www.researchgate.net/publication/283358689_A_behavioral_framework_for_dollar-cost_averaging_Dollar-cost_averaging_may_not_be_rational_behavior_but_it_is_perfectly_normal_behavior), Meir Statman, Journal of Portfolio Management Volume 22, pages 70-78, January 1995.
- Dollar Cost Averaging: A Behavioral View (https://blog.wealthfront.com/dollar-cost-averaging-behavioral-view/), Meir Statman, Wealthfront Blog, June 18, 2015.
- Dollar Cost Averaging (http://www.dimensional.com/famafrench/2009/06/dollar-cost-averaging.html), Fama/French Forum video.
- Does Dollar Cost Averaging Really Work (http://www.moneychimp.com/features/dollar_cost.htm), DCA calculator, moneychimp.
- Do Not Dollar-Cost-Average for More than Twelve Months (http://www.efficientfrontier.com/ef/997/dca.htm), Bill Jones, Efficient Frontier, October 1997.
- Dollar Cost Averaging (https://web.archive.org/web/20210812193032/http://www.sigmainvesting.com/advanced-investing-topics/dollar-cost-averaging), detailed examination by Sigma Investing.
- Bogleheads forum topic: "Finally! Can we lay DCA to rest and ban this topic forever?" (https://www.bogleheads.org/forum/viewtopic.php?t=83767). October 7, 2011.
- Invest now or temporarily hold your cash? (https://web.archive.org/web/20170215161450/https://personal.vanguard.com/pdf/ISGDCA.pdf), 2016 Vanguard paper comparing DCA to lump sum investing.
- Take the plunge, or drip by drip? (http://www.cbsnews.com/news/take-plunge-or-drip-by-drip/), CBS MoneyWatch article by Larry Swedroe, January 3, 2014.
- 6 Things To Consider When Investing a Lump Sum (https://www.forbes.com/sites/rickferri/2015/09/08/6-things-to-consider-when-investing-a-lump-sum/), Rick Ferri, Forbes, September 8, 2015.
- What if You Only Invested at Market Peaks? (https://awealthofcommonsense.com/2014/02/worlds-worst-market-timer/), by Ben Carlson on awealthofcommonsense.com (https://awealthofcommonsense.com).

Bibliography

- George M. Constantinides, "A Note on the Suboptimality of Dollar-Cost Averaging as an Investment Policy (https://web.archive.org/web/20190412180131/http://faculty.chicagobooth.edu/george.constantinides/documents/JFQA_1979.pdf)", Journal of Financial and Quantitative Analysis 14 (June 1979), 443-49.

- Wei Fang, "Dollar-cost Averaging: An Investigation (https://web.archive.org/web/20140123220801/http://www.valueaveraging.ca:80/docs/DCA%20Investigation%20Study.pdf)", The University of Nottingham, 2007
- Simon Hayley, "Value Averaging and the Automated Bias of Performance Measures (https://ssrn.com/abstract=1606347)", (March 11, 2011). Available at SSRN.
- Kim Johnson, and Tom Kruegar, "Market Timing versus Dollar-Cost Averaging: Evidence based on Two Decades of Standard & Poor's 500 Index Values (https://www.valueaveraging.ca/docs/Analysis_Dollar_Cost_Averaging.pdf)"
- John K. Knight, and Lewis Marshall, "Nobody Gains from DCA (https://web.archive.org/web/20140924161913/http://www.valueaveraging.ca/docs/Nobody%20Gains%20from%20DCA.pdf), Financial Services Review, 2(1): 51-61."
- Karyl B. Leggio, and Donald Lien, "Does loss aversion explain dollar cost averaging? (https://web.archive.org/web/20140916164735/http://valueaveraging.ca/docs/DCA%20Losses.pdf)", Financial Services Review, 10(2001): 117-127.
- Paul S. Marshall, "A Statistical Comparison Of Value Averaging vs. Dollar Cost Averaging and Random Investment Techniques (https://web.archive.org/web/20110814011724/http://www.studyfinance.com/jfsd/pdffiles/v13n1/marshall.pdf)", Journal of Financial and Strategic Decisions Volume 13, Number 1, Spring 2000.
- William J. Trainor Jr., "Within-horizon exposure to loss for dollar cost averaging and lump sum investing (https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.392.2893&rep=rep1&type=pdf)", Financial Services Review 14 (2005) 319–330.

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